Lakshya Jindal

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Education

International Institute of Information Technology, Hyderabad Aug 2023 – Present
B.Tech + MS (By Research) in Electronics and Communication Engineering *Hyderabad, India*

Hiranandani Foundation School Thane June 2021 – June 2023
ISC (Class XII), PCM and Computer Applications – 94.25% *Mumbai, India*

Experience

Undergraduate Researcher | Robotics Research Centre, IIIT Hyderabad April 2025 – Present
Research Assistant *On-site*

- Designed and implemented real-time motion planning and numerical optimization algorithms in **C++ and Python**, optimizing computational latency, memory footprint, and numerical convergence.
- Built modular simulation pipelines to benchmark algorithmic performance under stochastic sensor noise, focusing on **software reliability, maintainability, and systematic verification**.
- Collaborated with researchers to transition research prototypes into production-quality software architectures, conducting systematic profiling, automated testing, and regression analysis.

Projects

Signal Reconstruction & Audio Spectral Analysis | *DSP, STFT, Spline/Sinc Interpolation, K-Means* |

- Formulated signal reconstruction algorithms for non-uniform jittered sampling using **cubic spline interpolation** and missing-sample recovery via **Sinc interpolation**, evaluating MSE across varying jitter factors.
- Implemented an audio DSP pipeline using **STFT** and low-pass filtering; extracted spectral features (**Centroid, Bandwidth, 85% Roll-off**) to classify transient drum hits via **K-Means clustering**.

Quadrature Down-Converter (QDC) | *Analog Electronics, RF Circuits, LTSpice, Frequency Translation* | GitHub

- Designed and simulated a **100 kHz RF-to-IF quadrature down-converter** in LTSpice, modeling wireless communication receiver front-ends (**Wi-Fi, Bluetooth**).
- Integrated a **local oscillator,** 90° **phase shifter, analog mixer, and active low-pass filter** to generate in-phase (I) and quadrature (Q) baseband signals with high image rejection.

Multi-Stage Audio Amplifier Design | *Analog Signal Chain, Active Filters, Small-Signal Analysis, LTSpice* | GitHub

- Designed and built a **multi-stage analog amplifier** delivering > 1.5**W** to a 10Ω load (±5V supply), integrating a differential pre-amp, CE gain stage, **20 Hz–20 kHz band-pass filter**, and Class-AB power stage.
- Simulated frequency response, gain-bandwidth product, and THD in **LTSpice**; verified stability, SNR, and transient performance via **oscilloscope-based hardware measurements**.

RISC-V Processor Design | *SystemVerilog, RTL Design, Computer Architecture, Verification* | GitHub

- Architected single-cycle and **5-stage pipelined RISC-V processors** in **SystemVerilog** with **hazard detection, data forwarding, and branch resolution logic** to maximize instruction throughput.
- Verified execution correctness across data and control hazard test suites using hardware simulation.

Technical Skills

Signal Processing & DSP: Fourier Transforms (CTFT, DTFT, DFT/FFT, STFT), Sampling Theorem, Sinc/Spline Reconstruction, Digital & Active Filters, Spectral Analysis, System Classification & Convolution
Hardware & Circuits: Analog Signal Chain, Transimpedance Amplifiers, Differential & Class-AB Amplifiers, RF Mixers, I/Q Demodulation, CMOS, RTL Design, FPGA Prototyping
Languages & Tools: C, C++, Python (NumPy, SciPy), MATLAB, SystemVerilog, Verilog, SPICE/LTSpice, Linux, Git, Benchtop Oscilloscopes

Achievements/Extracurriculars

- Achieved a peak rating of **1269** on **Codeforces** (Handle: `Laksh_J`).
- Secured **3rd Place** in **Megathon 2025**, developing a multi-modal AI damage assessment pipeline for vehicles.
- Solved **200+** Data Structures and Algorithms problems across Codeforces, LeetCode, and CodeChef.
- Secured **AIR 4391** in JEE Mains 2023 and **AIR 10931** in JEE Advanced 2023 (Top 0.5% nationally).
- Qualified for **KVPY 2022** Fellowship and cleared **Pre-RMO 2019** (National Mathematics Olympiad Stage 1).

Relevant Coursework

Signals & Systems: Signal Processing, Networks Systems and Signals, Communication Theory, Probability & Random Processes
Circuits & Computing: Analog Electronic Circuits, VLSI Design, Intro to Processor Architecture, Data Structures and Algorithms